Lab 02 – Relational Model

Objectives:

The purpose of the lab of is to familiarize yourself with the User Interface, SQL Developer, and the database that we will be using throughout the course to communicate with the Oracle server. By the end of this lab, you should be able to:

- Successfully establish a connection with and login to the Oracle database server using SQL Developer
- Explore and work with the database and data
- Understand the relationships, constraints, data types, and tables' specification.

Preface:

If you have not already done so, you will need to download the sample database creation script from blackboard and run it. These instructions are included in the Getting Started section with SQL Developer document.

SUBMISSION

Answer the following questions in the provided space. **Save your file as a PDF file and name it as following**:

DBS211_L02_Group#.sql.

Tasks:

By navigating through SQL Developer and looking at the Columns, Data, model, and Constraints tabs for the given tables. You will answer the following questions.

NOTE: In Question (a), some questions are answered as examples. You need to complete the rest. Add more rows to the tables in the document if you need more space for an answer. Use a different color for your answers.

For the given tables in your database, answer the following questions:

Part A

See the sample question:

a) Answer the following Question for the **DBS211_PRODUCTS** table.

1) How many columns (attributes) are there in this table? _______9_______
2) How many rows are there in this table? ________110_________
3) List the table's columns and the requested information in the following format:

Column Name	**Type**	**Not Null**
PRODUCTCODE	VARCHAR2 (15 BYTE)	NOT NULL
PRODCUTNAME	VARCHAR2 (70 BYTE)	NOT NULL
PRODUCTLINE	VARCHAR2 (50 BYTE)	NOT NULL
PRODUCTSCALE	VARCHAR2 (10 BYTE)	NOT NULL
PRODUCTVENDOR	VARCHAR2 (50 BYTE)	NOT NULL
PRODUCTDESCRIPTION	VARCHAR2 (1000 BYTE)	NOT NULL
QUANTITYINSTOCK	NUMBER (38, 0)	NOT NULL
BUYPRICE	NUMBER (10, 2)	NOT NULL
MSRP	NUMBER (10, 2)	NOT NULL

4) Sort the data based on the third column in your table and write the data of the first row in the following format. To sort the data based on a column, right click on that column, and select “sort”. You can select the column that the data will be sorted based on it. (Make sure CHATACTER type values are enclosed in single quotes.)

Column name	Column Value
PRODUCTCODE	‘S18 1984’
PRODUCTNAME	‘1995 HONDA CIVIC’
PRODUCTLINE	‘CLASSIC CARS’
PRODUCTSCALE	1:18
PRODUCTVENDOR	‘MIN LIN DIECAST’
PRODUCTDESCRIPTION	‘THIS MODEL FEATURES, OPENING HOOD, OPENING DOORS, DETAILED ENGINE, REAR SPOILER, OPENING TRICK, WORKING STEERING, TINTED WINDOWS, BAKED NAMEL FINISH. COLOR YELLOW.’
QUANTITYINSTOCK	9772
BUYPRICE	93.89
MSRP	142.25

5) List all constraints in this table.
If a constraint is a foreign key, write the reference table.

Constraint Name	**Constraint Type**	**Constraint on Column**	**Constraint Condition**	**Reference Table**
DBS211_PROD_LINE_FK	Foreign_Key	PRODUCTLINE	(NULL)	DBS211_PRODUCTLINES
SYS_C002714816	Check	PRODUCTCODE	"PRODUCTCODE" IS NOT NULL	
SYS_C002714817	Check	PRODUCTNAME	"PRODUCTNAME" IS NOT NULL	
SYS_C002714818	Check	PRODUCTLINE	"PRODUCTLINE" IS NOT NULL	
SYS_C002714819	Check	PRODUCTSCALE	"PRODUCTSCALE" IS NOT NULL	
SYS_C002714820	Check	PRODUCTVENDOR	"PRODUCTVENDOR" IS NOT NULL	
SYS_C002714821	Check	PRODUCTDESCRIPTION	"PRODUCTDESCRIPTION" IS NOT NULL	
SYS_C002714822	Check	QUANTITYINSTOCK	"QUANTITYINSTOCK" IS NOT NULL	
SYS_C002714823	Check	BUYPRICE	"BUYPRICE" IS NOT NULL	

SYS_C002714824	Check	MSRP	"MSRP" IS NOT NULL	
SYS_C001034319	Primary_Key	PRODUCTCODE	(NULL)	

6) What tables are in relationship with this table? List them below.

Table Name	Column in Common
DBS211_ORDERSDETAILS	PRODUCTCODE
DBS211_PRODUCTLINES	PRODUCTLINE

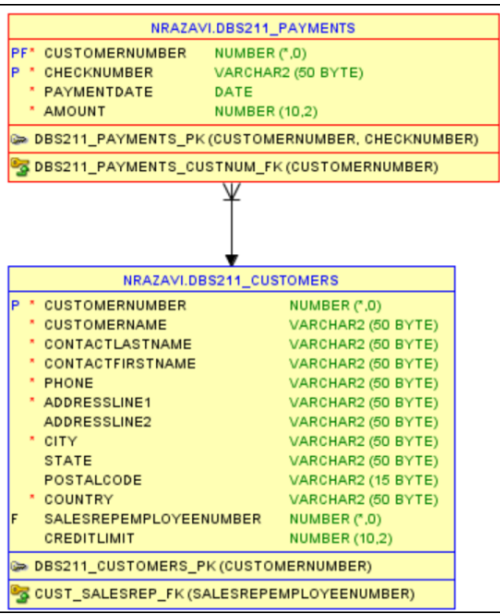
7) What is the model for this table relationships?

NOTE: means MANY

means ONE

MANY ( ) is close to Contacts. You read “many Contacts”.

ONE ( ) is close to customers. You read “one customer”.



8) Translate the relationships in Question 7 (model) to English.

In the relationship of the tables above, the CUSTOMER table has many payments.

b) Answer the following Question for the **DBS211_CUSTOMERS** table.

1) How many columns (attributes) are there in this table? ______13________
2) How many rows are there in this table? _________112__________
3) List the table’s columns and the requested information in the following format:

Column Name	Type	Not Null
CUSTOMERNUMBER	NUMBER(38,0)	NOT NULL
CUSTOMERNAME	VARCHAR2(50 BYTE)	NOT NULL
CONTACTLASTNAME	VARCHAR2(50 BYTE)	NOT NULL

CONTACTFIRSTNAME	VARCHAR2(50 BYTE)	NOT NULL
PHONE	VARCHAR2(50 BYTE)	NOT NULL
ADDRESSLINE1	VARCHAR2(50 BYTE)	NOT NULL
ADDRESSLINE2	VARCHAR2(50 BYTE)	NULL
CITY	VARCHAR2(50 BYTE)	NOT NULL
STATE	VARCHAR2(50 BYTE)	NULL
POSTALCODE	VARCHAR2(15 BYTE)	NULL
COUNTRY	VARCHAR2(50 BYTE)	NOT NULL
SALESREPEMPLOYEENUMBER	NUMBER(38,0)	NULL
CREDITLIMIT	NUMBER(10,2)	NOT NULL

4) Sort the data based on the third column in your table and write the data of the first row in the following format: (Make sure **CHATACTER** type values are enclosed in 'single quotes'.)

Column Name	Column Value
CUSTOMERNUMBER	249
CUSTOMERNAME	'Amica Models "&" Co.'
CONTACTLASTNAME	'Accorti'
CONTACTFIRSTNAME	'Paolo'
PHONE	'011-4988555'
ADDRESSLINE1	'Via Monte Bianco 34'
ADDRESSLINE2	(NULL)
CITY	'Torino'
STATE	(NULL)
POSTALCODE	'10100'
COUNTRY	'Italy'
SALESREPEMPLOYEENUMBER	1401
CREDITLIMIT	113000

5) List all constraints in this table.
If a constraint is a foreign key, write the reference table.

Constraint Name	Constraint Type	Constraint on Column	Constraint Condition	Reference Table
CUST_SALESREP_FK	Foreign_Key	SALESREPEMPLOYEENUMBER	(null)	EMPLOYEES
SYS_C002714804	Check	CUSTOMERNUMBER	"CUSTOMERNUMBER" IS NOT NULL	
SYS_C002714805	Check	CUSTOMERNAME	"CUSTOMERNAME" IS NOT NULL	
SYS_C002714806	Check	CONTACTLASTNAME	"CONTACTLASTNAME" IS NOT NULL	
SYS_C002714807	Check	CONTACTFIRSTNAME	"CONTACTFIRSTNAME" IS NOT NULL	

SYS_C002714808	Check	PHONE	"PHONE" IS NOT NULL	
SYS_C002714809	Check	ADDRESSLINE1	"ADDRESSLINE1" IS NOT NULL	
SYS_C002714810	Check	CITY	"CITY" IS NOT NULL	
SYS_C002714811	Check	COUNTRY	"COUNTRY" IS NOT NULL	
SYS_C002714812	Primary_Key	CUSTOMERNUMBER	(null)	

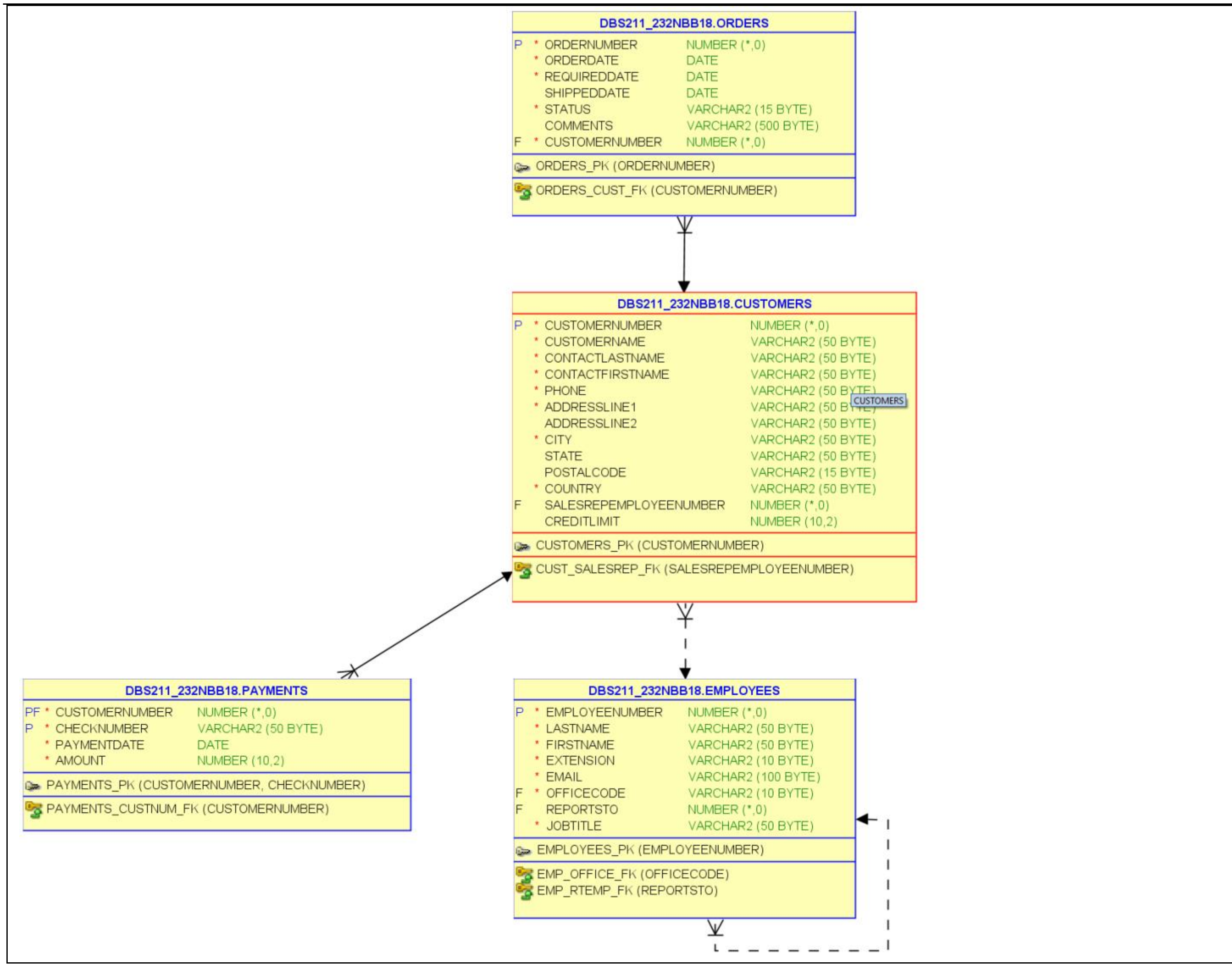
6) What tables are in relationship with this table? List them below.

Table Name	**Column in Common**
ORDERS	CUSTOMERNUMBER
EMPLOYEES	SALESREPEMPLOYEESNUMBER
PAYMENTS	CUSTOMERNUMBER

7) What is the model for this table relationships?

NOTE: means MANY

means ONE



8) Translate all the relationships in Question 7 (model) to English.

The model above shows that there are many orders have a singular customer. Therefore, one employee has many customers. The payments table on the left shows that many payments have one customer.

MANY ORDERS HAVE ONE CUSTOMER
ONE EMPLOYEE IS ASSIGNED TO MANY CUSTOMERS
MANY PAYMENTS ARE MADE BY A CUSTOMER

c) Answer the following Question for the **DBS211_EMPLOYEES** table.

1) How many columns (attributes) are there in this table? ______8________
2) How many rows are there in this table? _______23__________

3) List the table’s columns and the requested information in the following format:

Column Name	Type	Not Null
EMPLOYEENUMBER	NUMBER(38,0)	NOT NULL
LASTNAME	VARCHAR2(50 BYTE)	NOT NULL
FIRSTNAME	VARCHAR2(50 BYTE)	NOT NULL
EXTENSION	VARCHAR2(10 BYTE)	NOT NULL
EMAIL	VARCHAR2(100 BYTE)	NOT NULL
OFFICECODE	VARCHAR2(10 BYTE)	NOT NULL
REPORTSTO	NUMBER(38,0)	NULL
JOBTITLE	VARCHAR2(50 BYTE)	NOT NULL

4) Sort the data based on the third column in your table and write the data of the first row in the following format: (Make sure **CHATACTER** type values are enclosed in single quotes.)

Column Name	Column Value
EMPLOYEENUMBER	1161
LASTNAME	'Fixter'
FIRSTNAME	'Andy'
EXTENSION	'X101'
EMAIL	'afixter@classicmodelcars.com'
OFFICECODE	'6'
REPORTSTO	1088
JOBTITLE	'Sales Rep'

5) List all constraints in this table.
If a constraint is a foreign key, write the reference table.

Constraint Name	Constraint Type	Constraint on Column	Constraint Condition	Reference Table
EMP_OFFICE_FK	Foreign_Key	OFFICECODE	(null)	OFFICES
EMP_RTEMP_FK	Foreign_Key	REPORTSTO	(null)	EMPLOYEES
SYS_C002714794	Check	EMPLOYEENUMBER	"EMPLOYEENUMBER" IS NOT NULL	
SYS_C002714795	Check	LASTNAME	"LASTNAME" IS NOT NULL	
SYS_C002714796	Check	FIRSTNAME	"FIRSTNAME" IS NOT NULL	
SYS_C002714797	Check	EXTENSION	"EXTENSION" IS NOT NULL	
SYS_C002714798	Check	EMAIL	"EMAIL" IS NOT NULL	
SYS_C002714799	Check	OFFICECODE	"OFFICECODE" IS NOT NULL	
SYS_C002714800	Check	JOBTITLE	"JOBTITLE" IS NOT NULL	
SYS_C002714801	Primary_Key	EMPLOYEENUMBER	(null)	

6) What tables are in relationship with this table? List them below.

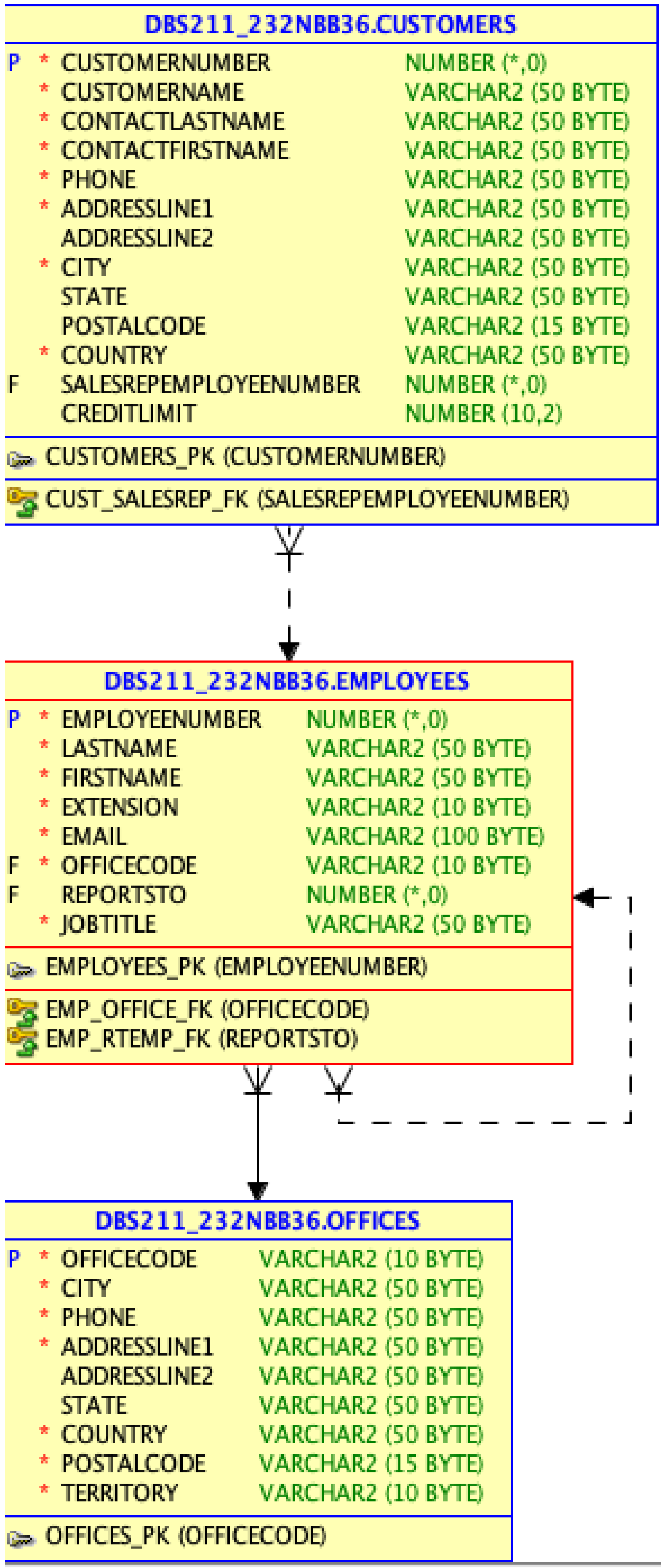
Table Name	Column in Common

CUSTOMERS	EMPLOYEENUMBER
OFFICES	OFFICECODE

7) What is the model for this table relationships?

NOTE: means MANY

↓ means ONE


DBS211_232NBB36.CUSTOMERS
P * CUSTOMERNUMBER NUMBER (*,0)
* CUSTOMERNAME VARCHAR2 (50 BYTE)
* CONTACTLASTNAME VARCHAR2 (50 BYTE)
* CONTACTFIRSTNAME VARCHAR2 (50 BYTE)
* PHONE VARCHAR2 (50 BYTE)
* ADDRESSLINE1 VARCHAR2 (50 BYTE)
ADDRESSLINE2 VARCHAR2 (50 BYTE)
* CITY VARCHAR2 (50 BYTE)
STATE VARCHAR2 (50 BYTE)
POSTALCODE VARCHAR2 (15 BYTE)
* COUNTRY VARCHAR2 (50 BYTE)
F SALESREPEMPLOYEENUMBER NUMBER (*,0)
CREDITLIMIT NUMBER (10,2)
CUSTOMERS_PK (CUSTOMERNUMBER)
CUST_SALESREP_FK (SALESREPEMPLOYEENUMBER)
DBS211_232NBB36.EMPLOYEES
P * EMPLOYEENUMBER NUMBER (*,0)
* LASTNAME VARCHAR2 (50 BYTE)
* FIRSTNAME VARCHAR2 (50 BYTE)
* EXTENSION VARCHAR2 (10 BYTE)
* EMAIL VARCHAR2 (100 BYTE)
F * OFFICECODE VARCHAR2 (10 BYTE)
F REPORTSTO NUMBER (*,0)
* JOBTITLE VARCHAR2 (50 BYTE)
EMPLOYEES_PK (EMPLOYEENUMBER)
EMP_OFFICE_FK (OFFICECODE)
EMP_RTEMP_FK (REPORTSTO)
DBS211_232NBB36.OFFICES
P * OFFICECODE VARCHAR2 (10 BYTE)
* CITY VARCHAR2 (50 BYTE)
* PHONE VARCHAR2 (50 BYTE)
* ADDRESSLINE1 VARCHAR2 (50 BYTE)
ADDRESSLINE2 VARCHAR2 (50 BYTE)
STATE VARCHAR2 (50 BYTE)
* COUNTRY VARCHAR2 (50 BYTE)
* POSTALCODE VARCHAR2 (15 BYTE)
* TERRITORY VARCHAR2 (10 BYTE)
OFFICES_PK (OFFICECODE)

8) Translate all the relationships in Question 7 (model) to English.

Here the customers have many employees, whereas many employees have one office. One EMPLOYEE HAVE MAY CUSTOMERS, One office has many EMPLOYEES. Many employees reports to one EMPLOYEE.

d) Answer the following Question for the **DBS211_ORDERS** table.

1) How many columns (attributes) are there in this table? ______7________
2) How many rows are there in this table? _______103__________
3) List the table's columns and the requested information in the following format:

Column Name	Type	Not Null
ORDERNUMBER	NUMBER(38,0)	NOT NULL
ORDERDATE	DATE	NOT NULL
REQUIREDDATE	DATE	NOT NULL
SHIPPEDDATE	DATE	NULL
STATUS	VARCHAR2(15 BYTE)	NOT NULL
COMMENTS	VARCHAR2(500 BYTE)	NULL
CUSTOMERNUMBER	NUMBER(38,0)	NOT NULL

4) Sort the data based on the third column in your table and write the data of the first row in the following format: (Make sure **CHATACTER** type values are enclosed in single quotes.)

Column Name	Column Value
ORDERNUMBER	10100
ORDERDATE	'03-01-06'
REQUIREDDATE	'03-01-13'
SHIPPEDDATE	'03-01-10'
STATUS	'Shipped'
COMMENTS	(null)
CUSTOMERNUMBER	363

5) List all constraints in this table.
If a constraint is a foreign key, write the reference table.

Constraint Name	Constraint Type	Constraint on Column	Constraint Condition	Reference Table
ORDERS_CUST_FK	Foreign_Key	CUSTOMERNUMBER	(null)	CUSTOMERS
SYS_C002714827	Check	ORDERNUMBER	"ORDERNUMBER" IS NOT NULL	
SYS_C002714828	Check	ORDERDATE	ORDERDATE	
SYS_C002714829	Check	REQUIREDDATE	"REQUIREDDATE" IS NOT NULL	

SYS_C002714830	Check	STATUS	"STATUS" IS NOT NULL	
SYS_C002714831	Check	CUSTOMERNUMBER	"CUSTOMERNUMBER" IS NOT NULL	
SYS_C002714832	Primary_Key	ORDERNUMBER	(null)	

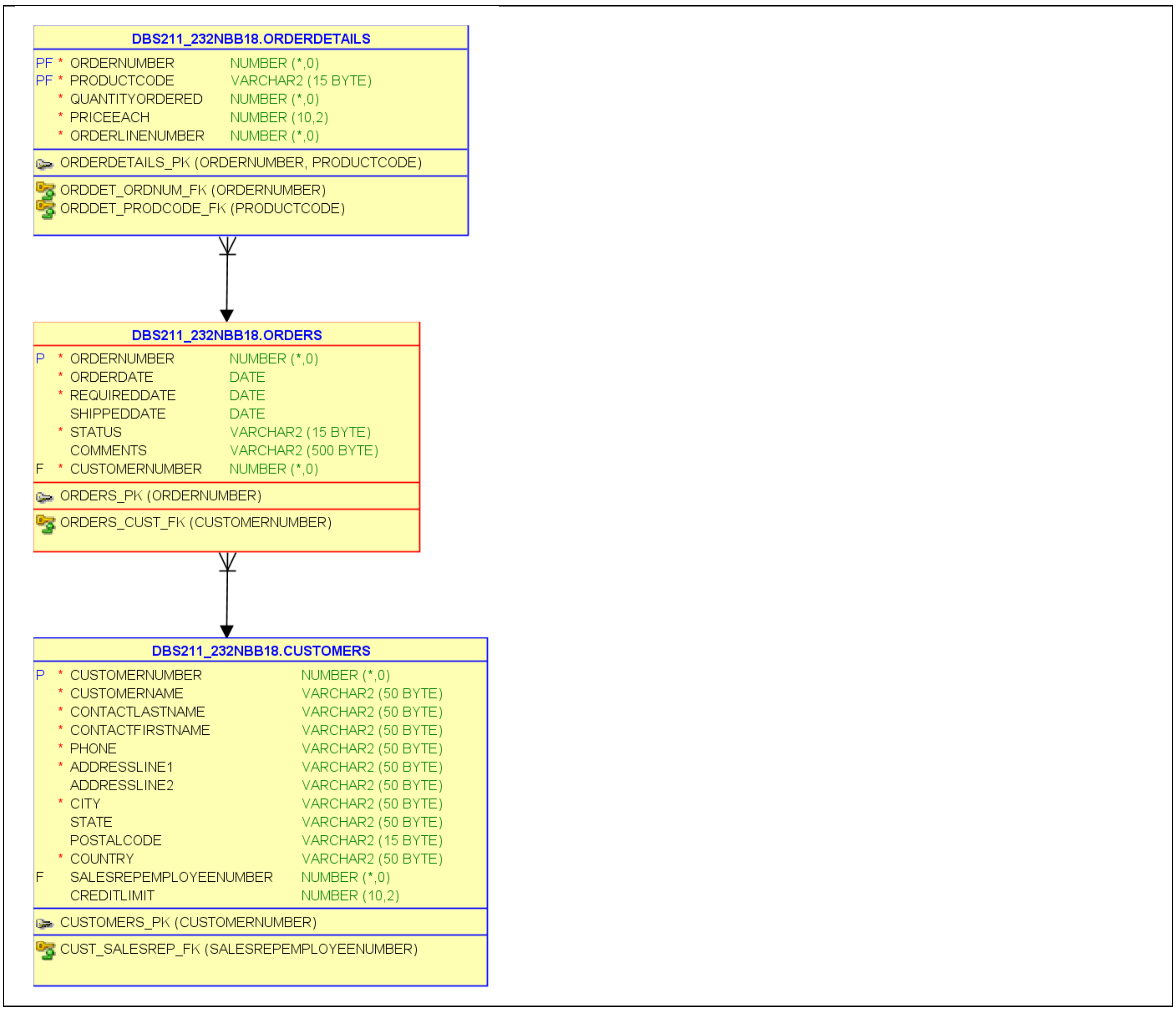
6) What tables are in relationship with this table? List them below.

Table Name	Column in Common	Refers to
ORDERSDETAILS	ORDERNUMBER	
CUSTOMERS	CUSTOMERNUMBER	

7) What is the model for this table relationships?

NOTE: means MANY

means ONE



8) Translate all the relationships in Question 7 (model) to English.

There are many ORDERDETAILS for ORDERS, by extension, there are many ORDERS for CUSTOMERS.
One CUSTOMER has many ORDERS.
One ORDER has many ORDERDETAILS.

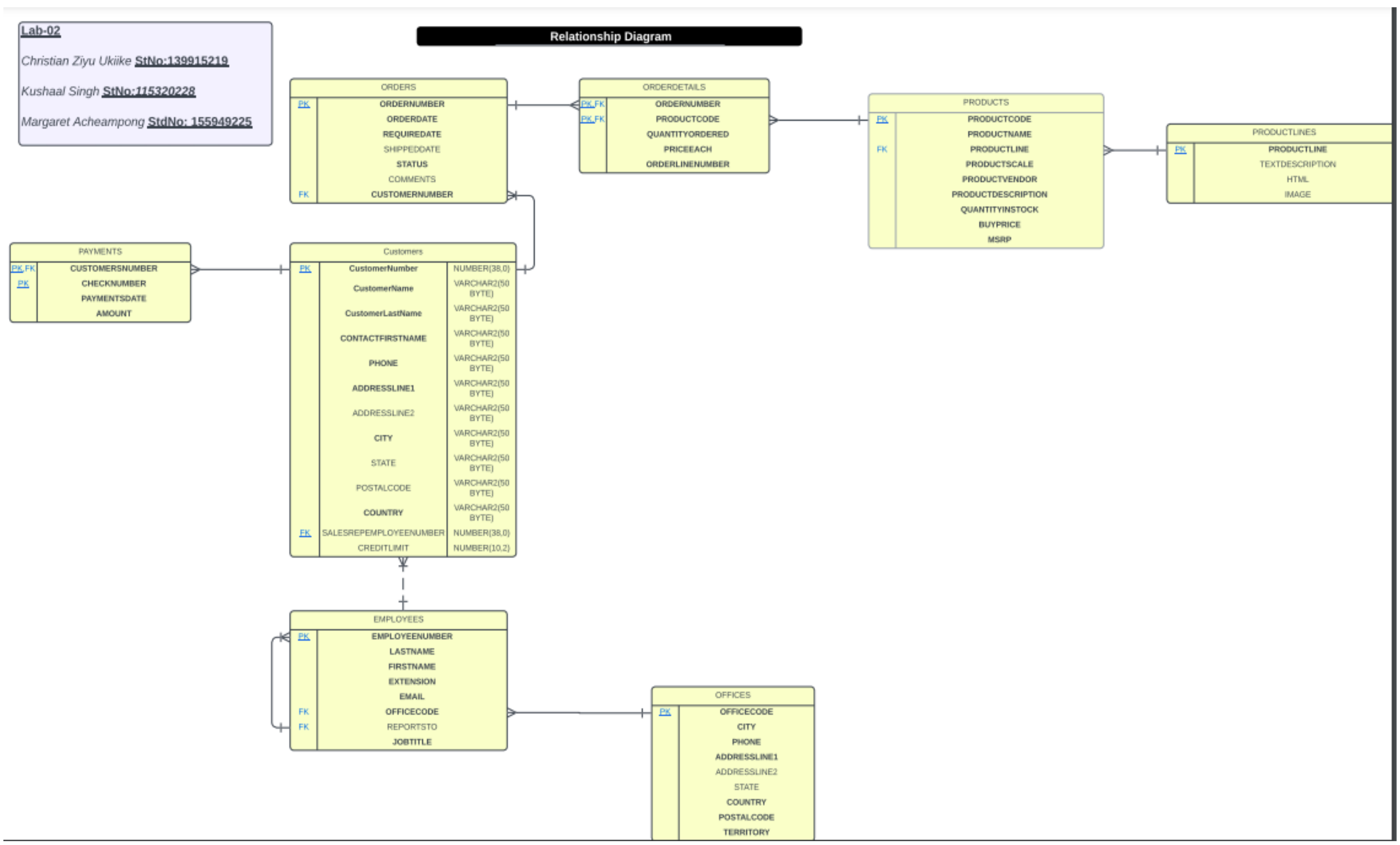
Part B

Create a relationship diagram for all the tables in the database. Use the MODEL tab to see the tables (entities) and their relationships.

Your diagram must include:

- All 8 tables
- The names of the entities (tables)
- The attributes (columns) for each table
- Lines representing the relationships between tables
- Crows Foot Symbols on the lines representing the type of relationship (1-1, 1-many)
- Required fields should be bolded
- Primary Key fields should be underlined **or** indicated with a PK beside it.
- Child fields in the relationships should be indicated with an FK beside it.

Use Lucid chart to draw you diagram. Save the diagram as an image and insert it here in the following box.



Good Luck.